

[NJC/2024/Q1]

1 Name your Jupyter Notebook as:

TASK1_<your name>.ipynb

The task is to implement a Vigenère cypher encryption algorithm via the use of Caesar cypher.

A Caesar cypher encodes each letter with a different letter. A n-place Caesar cypher uses the ASCII value of each letter and adds the number n , where $1 \leq n \leq 26$ to it.

For example, if $n = 10$,

- The letter 'A' has the ASCII value 65. Adding 10 to the number will give the ASCII value 75. The character for 75 is 'K'.
- When an uppercase letter's code goes beyond 'Z' it returns to 'A'. For example, the character 'Z' will be encrypted as 'J'.
- When a lowercase letter's code goes beyond 'z' it returns to 'a'. For example, the character 'x' will be encrypted as 'h'.
- Spaces ' ' are replaced with the character '!'.

You will only need to convert letters and spaces. If the character is invalid, -1 is returned.

[NJC/2024/Q1 Cont']

The following table shows the ASCII values of some of the characters.

Character	ASCII value
A	65
Z	90
a	97
z	122
"(space)	32
!	33

In ASCII the letters follow on numerically. For example, the letter 'A' is 65, 'B' is 66, 'C' is 67 etc.

For each of the sub-tasks, add a comment statement at the beginning of the code, using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

In [1]:

#Task 1.1

Program Code

Output:

Task 1.1

Write program code for a function task1_1() that:

- takes two parameters:
 - a character c, and
 - an integer n.
- encrypt the character c by adding the number n, where $1 \leq n \leq 26$, to its ASCII value.
- Spaces will be replaced with '!'.
- returns the new encrypted character. If the character is invalid, -1 is returned. [5]

[NJC/2024/Q1 Cont']

Task 1.2

The Vigenère cypher is a method of encrypting alphabetic text where each letter of the plaintext is encoded with a different Caesar cipher, whose increment is determined by the corresponding letter of another text, the keyword key.

For example, suppose that the plaintext to be encrypted is 'Attack#at dawn' and the keyword is 'LEMON'.

The keyword would be repeated until it matches the length of the plaintext.

Plaintext : Attack#at dawn

Key : LEMONLEMONLEMO

Ciphertext: Mygpqw#ni!pfjc

Because the 'L' is the twelfth letter in the alphabet, the first letter in the message ('A') is replaced by the letter which comes twelve places after it in the alphabet ('M'). In other words, 'A' is encrypted with 12-place Caesar cypher.

Because the 'E' is the fifth letter in the alphabet, the second letter in the message ('t') is replaced by the letter which comes five places after it in the alphabet ('y'). In other words, 't' is encrypted with 5-place Caesar cypher.

Because '#' is not an alphabet, it is an invalid character and doesn't get encrypted.

And so on, until the complete message has been encrypted.

The text file TASK1DATA.txt contains a message that needs to be encrypted with the keyword 'Njc' and then stored in a text file named ENCRYPTEDMESSAGE.txt

[NJC/2024/Q1 Cont']

Write program code to:

- read the data from the text file TASK1DATA.txt
- encrypt each character with the Vigenère cypher and keyword 'Njc'
- store the encrypted message in the text file ENCRYPTEDMESSAGE.txt

Keep the **invalid** characters as it is when appending to the encrypted message. [8]

Test your program with TASK1DATA.txt

Show the contents of ENCRYPTEDMESSAGE.txt after you have run the program. [1]

Save your Jupyter Notebook for Task 1.